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Prediction of retrogressive landslide in sensitive clays by incorporating a novel strain softening law into the Material Point Method

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ABSTRACT

Sensitive clays, when subjected to large strains, exhibit a unique strain-softening behavior, transforming into a remolded liquid with remarkably low shear strength. When a slope fails, this behavior leads to the remolded clay moving away from its original position, triggering subsequent failures and catastrophic outcomes. To accurately predict such scenarios, it is crucial to incorporate realistic strain-softening characteristics into the constitutive soil model. This paper presents a novel yet practical strain-softening law developed by the authors that effectively captures the post-peak behavior of sensitive clays down to their remolded strength. The softening law is implemented in an elastoplastic Mohr-Coulomb model and incorporated into Anura3D, an open-source software that uses the Material Point Method to simulate large deformations. The constitutive model is calibrated by simulating the stress-strain behavior through direct shear tests conducted at three sensitive clay landslide locations. The accuracy of the overall numerical framework is assessed by predicting the post-failure movements of these landslides. Notably, two of the landslides, Sainte-Monique (1994) and Saint-Jude (2010), have previously been analyzed using other numerical tools, allowing for a comparative analysis with the method presented here. The third landslide, the Saint-Luc-de-Vincennes landslide, representing a composite flow slide and spread, has been numerically simulated for the first time. The post-failure behavior observed in the landslide events is compared with field observations and other numerical analyses. The results show that the MPM models with the proposed strain-softening law can reasonably predict post-failure retrogression and runout distance, which are crucial parameters for determining the risk of landslides in sensitive clays.

1. Introduction

Large retrogressive landslides in sensitive clays frequently occur in Scandinavia and eastern Canada. Previous studies have identified the primary cause of these landslides as the significant reduction in post-peak shear strength, commonly referred to as "strain-softening," in these soils (Jin et al., 2020a; Shen et al., 2024; Wang et al., 2022). This strain-softening nature can be traced back to their depositional history (Urmi et al., 2023c). Sensitive clays were deposited in depressions created by the retreat of the Laurentian ice sheet approximately 14,000 to 6000 years ago in a marine environment. During this process, the clays acquired a flocculated soil structure characterized by high undisturbed shear strength from exposure to salt ions. However, due to the isostatic rebound, these clay deposits were uplifted above sea level and exposed to fresh water. Consequently, removing salt ions from the pores to values of about 2–5 g/l can result in a metastable sensitive clay structure (Lefebvre, 2017). This metastable structure is destroyed upon remolding, and the liberated interparticle surface water transforms the clay into a liquid-like fluid, commonly referred to as remolded clay. During this process, the shear strength reduces from peak (s_{up}) to remolded (s_{ur}) strength, and the sensitivity of the soil (S_t) is quantified as the ratio between peak and remolded shear strengths (s_{up}/s_{ur}) (Skempton and Northey, 1952). When subjected to loading, the potential to liquefy or get remolded is one of the leading agents governing the post-failure behavior of sensitive clays. When excessive stress surpasses their peak shear strength, narrow shear bands with localized strain form, triggering the structural transformation that remolds the soil mass. The remolded clays within the shear bands can either spread laterally or drift away from their original position, resulting in a series of failures that progress in either a progressive or retrogressive manner (Urmi et al.,

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2023a). These landslides can extend over kilometers, resulting in widespread destruction and devastating aftermath in the affected areas. The catastrophic consequences of landslides occurring in sensitive clays have sparked significant interest among researchers who have extensively investigated the behavior of these materials under different loading conditions (Gylland et al., 2013; Kennedy et al., 2021; Lefebvre, 2017; Okamoto et al., 2004). The accurate prediction of the post-failure mechanism and the potentially affected area can greatly facilitate the evaluation of disaster risk and the development of effective mitigation measures (Troncone et al., 2023a; Urmi et al., 2023b).

The mechanism behind retrogressive failure in sensitive soil involves various complex features, including the landslide trigger, the formation of shear bands or multiple failure surfaces, the movement of remolded soil, and the progression of the landslide (Urmi et al., 2023a). These complexities pose significant challenges when conducting numerical analyses of sensitive clay landslides. Traditional methods, such as limit equilibrium or finite element strength reduction techniques, face two major issues that render them unsuitable for application in sensitive clay slopes. Firstly, these conventional approaches have limited capability in predicting landslide initiation and cannot anticipate subsequent failures or runouts in retrogressive landslides. Secondly, the large deformations associated with these landslides can cause mesh tangling in conventional finite element methods (Mohammadi and Taiebat, 2016). The strain-softening behavior exhibited by sensitive clays, responsible for the intricate failure mechanism, necessitates using sophisticated constitutive soil models and advanced numerical tools capable of handling large deformation problems (Urmi et al., 2022).

Over the past two decades, considerable progress has been made in addressing large deformation problems in geotechnical engineering. Various mesh-based and meshless numerical frameworks, such as the Arbitrary Lagrangian Finite Element Method, the Particle Finite Element Method, and the Material Point Method, have been successfully implemented to capture key characteristics of large retrogressive failures (Jin et al., 2020a; Shan et al., 2021; Tran and Sołowski, 2019; Wang et al., 2022). Additionally, the development of strain-softening constitutive soil models has further enhanced the ability to model these phenomena effectively. In recent studies, highly non-linear post-peak strain-softening of sensitive clays has inspired the implementation of an exponential strength degradation equation (Eq. 1) to capture the post-peak soil behavior (Tran and Sołowski, 2019; Wang et al., 2022; Yuan et al., 2020).

$$s_u = s_{ur} + (s_{up} - s_{ur})e^{-3\gamma/\gamma_{95}} \tag{1}$$

where s_u is the degraded shear strength after the peak, γ is the strain corresponding to s_u, and γ_{95} is the strain when the strength is reduced by 95% of the supposed total reduction ($s_{up} - s_{ur}$). Even though 95% strength reduction does not correspond to a wholly remolded state, the strain value at 95% strength reduction is so large that it cannot be attained with conventional laboratory shear tests (Locat et al., 2015, 2017). To effectively model sensitive clay landslides using the strain-softening equation, one requires the value of the large deformation parameter γ^{95}, which represents the strain value at 95% strength reduction. However, on a laboratory scale, shear tests can reach the strain value, corresponding to at most 40–50% strength reduction with DSS tests and 25–30% with Triaxial tests. This limitation makes it difficult to obtain γ^{95} directly from laboratory tests. Due to the unavailability of γ^{95}, researchers resort to the back analysis method to model sensitive clay landslides. They start with an assumed value of γ^{95}, later, this parameter is adjusted to match the simulation outcome with field observations.

To eliminate the need for back analysis and improve the numerical modeling of sensitive clay landslides, Urmi et al. (2023b) developed an empirical/ indirect method to determine the large deformation parameter with available laboratory and field investigation results. In the development process, it was found that the strain-softening equation requires modification based on site-specific data. The strain-softening patterns of the 9 prehistoric sensitive clay landslide sites revealed that the exponent of the strain-softening curve ($-3/\gamma_{95}$ in eq. 1) varies from site to site and isn't constant regarding its relation to the large deformation parameter, γ^{95}. Moreover, the strength reduction is not entirely exponential; at greater strain values, the exponent of the exponential curve reduces to zero; thus, the strength reduction becomes linear. These observations inspired Urmi et al. (2023b) to propose a new site-specific strain-softening equation (Eq. 2) based on the strain-softening pattern observed in the sensitive clays of prehistoric landslide locations. Based on the unique non-linear strain-softening behavior each site, it was observed that the exponential strength decrease required two site-specific parameters (α, β) for a better representation. These parameters are derivable from shear stress-strain curves from laboratory shear test results specific to the site in question. Direct Simple Shear (DSS) tests are preferred because they better represent the large deformation strain-softening behavior for progressive failure than triaxial tests (Locat et al., 2015). Urmi et al. (2023b) demonstrated that the exponential behavior characterized by site-specific parameters (α, β), derived from a laboratory-scale partial stress-strain curve, closely approximates the actual behavior of sensitive clays at high strain levels. The crucial aspect is identifying the pattern of exponential decay; even when derived from a limited number of data points on a partial curve, the results remain comparable because the strength decrease adheres to a unique law for a particular site.

The proposed softening equation is a combination of an exponential and linear strength reduction as represented by eq. 2(Fig. 1).

$$s_u = \begin{cases} s_{uR} + (s_{up} - s_{uR})\alpha e^{-\beta\gamma}, & s_u \geq s_{uR} \\ s_{uR} - (s_{uR} - s_{ur})(\gamma - \gamma_R)/(\gamma_r - \gamma_R), & s_{uR} > s_u \geq s_{ur} \end{cases} \tag{2}$$

Where, s_{uR} is the residual strength defined as the degraded strength up to which the strength reduction is exponential and can be determined by

$$s_{uR} = 0.095 s_{up} \tag{3}$$

γ_R is the corresponding residual strain at s_{uR}. γ_r is the strain at the remolded shear strength s_{ur} that can be estimated as,

$$\gamma_r = 0.13 s_{up}\gamma_R \tag{4}$$

Using eq. 2 facilitates the determination of the strain value at remolded shear stress (γ_r). The strain value at the remolded shear stress is particularly important because it completes the strength reduction curve and thus facilitates the evaluation of dissipated energy during the remolding, which is the area under the stress-strain curve. For more detailed information on the development process of eq. 2 and its application to landslide risk analysis, readers are encouraged to read Urmi et al. (2023b).

In this study, the strength of the sensitive clay was modeled using a non-associated Mohr-Coulomb constitutive model with an additional strain-softening law described by Eq. 2, where the reduction in cohesion

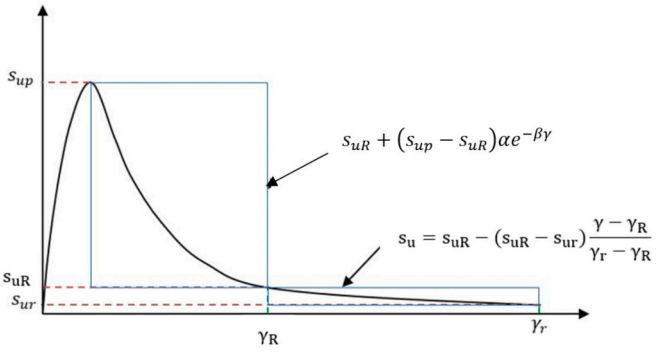


Fig. 1. Strain-softening behavior of sensitive clays (after Urmi et al., 2023b).

is equivalent to the degradation of undrained shear strength. For retrogressive failure analysis under undrained conditions, the friction angle in the Mohr-Coulomb model is set to zero, making it comparable to the Tresca softening model. Although this is a basic constitutive model, the incorporation of strain softening law on the Tresca or Von Mises criteria has been frequently used in simulating sensitive clay landslides under undrained conditions (Shan et al., 2021; Tran and Sołowski, 2019; Urmi et al., 2024; Wang et al., 2022; Yuan et al., 2020).

The primary objective of this paper is to validate the effectiveness of the new strain-softening equation to simulate sensitive clay landslides by implementing it in the Anura3D software, which utilizes a Material Point Method (MPM) framework (Anura3D MPM Research Community, 2022). The choice of MPM is motivated by its previous successful use in demonstrating its capabilities to handle complex problems in landslide analyses (Conte et al., 2020; González Acosta et al., 2021; Tran and Sołowski, 2019; Troncone et al., 2022a). This objective is achieved through the following steps. Firstly, direct shear tests were simulated to validate the implementation of the strain-softening equation in a user-defined constitutive soil model in Anura3D. This step evaluates the numerical model's reliability in predicting site-specific post-peak stress-strain behavior of sensitive clays. Next, the constitutive parameters of the softening equation were calibrated using the smeared crack approach through direct shear tests with different mesh sizes to mitigate the effect of mesh dependency for implementing strain-softening in a continuum numerical framework. The model was then validated by predicting the post-failure movement of three previously occurred sensitive clay landslides. A comparison was then conducted between the predicted post-failure behavior and the observed behavior in the field for all three case studies, allowing for insights into the use of numerical analysis in understanding the underlying mechanism of such failures.

This study focuses on three distinct landslide events in Quebec, Canada: the Sainte-Monique landslide in 1994, the Saint-Jude landslide in 2010, and the Saint-Luc-de-Vincennes landslide in 2016. It leverages the available numerical simulations of the Sainte-Monique and Saint-Jude landslides for comparative analysis and testing the effectiveness of the employed numerical framework. The Saint-Luc-de-Vincennes landslide, a composite flow slide and spread, presents a unique challenge and opportunity: it has never been previously modeled. Therefore, this study is the first one to apply the proposed constitutive model to this complex event, assessing its suitability in capturing the dual nature of the landslide; it highlights the need for innovative approaches in the analysis of retrogressive failure in sensitive clays.

2. Basis of MPM framework

The Material Point Method (MPM) represents a computational methodology initially formed by Harlow as the Particle-In-Cell method (PIC) at Los Alamos National Laboratory (York Ii et al., 1999). Primarily designed for fluid dynamics, the methodology was later expanded by Sulsky et al. (1994) at New Mexico University to accommodate solid mechanics challenges, specifically discretizing the dynamic momentum balance equation (de Vaucorbeil et al., 2020). Positioned between particle-based techniques and the Finite Element Method (FEM), MPM employs a dual discretization approach. In the first frame, the continuum undergoes discretization into material points (MPs), each signifying a sub-domain with an invariant mass to ensure conservation (Fig. 2). The classical MPM assumes mass concentration at the respective MP, encapsulating velocities, strains, and stresses. These MPs exhibit Lagrangian characteristics as they move with the deformations of the body. The second frame involves a computational mesh like the conventional FEM, encompassing the complete problem domain. Discrete governing equations are solved at mesh nodes, and variables are mapped from MPs to mesh nodes using mapping functions, typically linear shape functions. Boundary conditions may be imposed at mesh nodes or MPs, and the governing equations are incrementally solved. Finally, Material Point (MP) quantities are updated by interpolating mesh results without storing mesh information permanently. Not storing this information allows the mesh to be redefined at the end of each time step, although conventionally, the mesh stays the same.

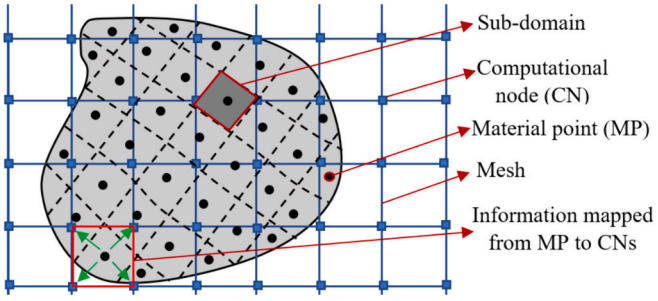


Fig. 2. Spatial formulation of MPM framework (after Urmi et al., 2023c).

The MPM approach provides a robust framework for analyzing complex geotechnical problems. Through its unique formulation and data transfer mechanism between MPs and computational nodes, the MPM technique can accurately simulate the behavior of slopes and other geotechnical systems, capturing their dynamic responses and interactions (e.g., Alsardi and Yerro, 2023; Yerro et al., 2016; Yost et al., 2022).

3. Strain-softening model calibration: simulation of direct shear tests

Locat et al. (2017, 2015, 2013) and Durand (2016) conducted multiple direct simple shear (DSS) tests using soil samples taken from the sensitive clay layers at three sites affected by previous landslides, i.e., Sainte-Monique, Saint-Jude, and Saint-Luc-de-Vincennes. They aimed to study the shear behavior of the clay involved in such landslides. Soil samples, with dimensions of 14 mm in height and 67.8 mm in diameter, were subjected to consolidation under a vertical stress level that closely matched the in-situ conditions corresponding to the depths from which the samples were obtained. The samples were placed inside plastic rings to prevent lateral deformation during consolidation. Shearing was conducted laterally by a motor, applying a horizontal strain rate of 0.065 mm/min at the top of the sample while the base remained fixed.

These tests were replicated in Anura3D, considering loading and boundary conditions consistent with the laboratory tests, as shown in Fig. 3. In particular, plain strain conditions were considered, and an initial rectangular domain (with a height equal to the sample's height as measured in laboratory testing) was divided into triangular elements (base length = height = 0.014 m) discretized with 46 MPs each. A total stress analysis is performed in undrained conditions. A prescribed velocity of 12 cm/min, consistent with the horizontal strain rate used in the laboratory shear test, was applied at the top of the row of nodes to generate the DSS total stress path. Input material parameters for the

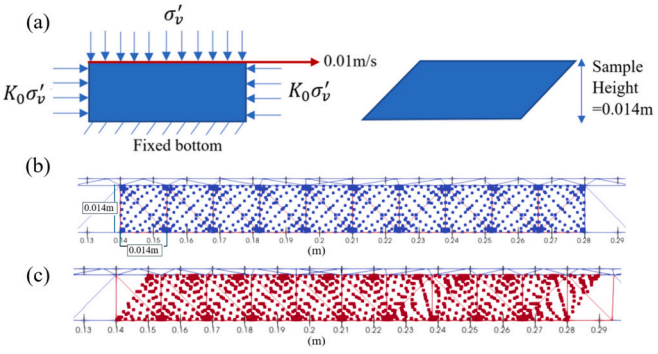


Fig. 3. (a) Loading conditions; configuration of material points (b) initial and (c) final.

simulations are listed in Table 1. Soil properties are obtained from the respective sites' detailed field investigation reports by Locat et al. (2017, 2015, 2013) and Tremblay-Auger et al. (2021). Specific parameters (s_{uR}, $\alpha, \beta, \gamma_R, \gamma_r$) for the softening equation (Eq.2) are based on laboratory test results obtained with a detailed methodology described in (Urmi et al., 2023b). The earth-pressure coefficient and Poisson's ratio are adopted from previous studies that have conducted numerical analyses specifically on the sites under consideration (Locat et al., 2017; Tran and Sołowski, 2019; Troncone et al., 2023b). The shear modulus of Eastern Canadian sensitive clays is typically lower than that of other clay soils, with values ranging from approximately $G \cong 30s_{up}$ to $100s_{up}$ (Quinn et al., 2011). For this study, the shear modulus is directly interpreted from the shear stress-strain curve obtained from the laboratory DSS test for each site, assuming that the behavior is elastic up to peak shear strength (Shear modulus, $G=s_{up}/\gamma_p$).

Figure 4 shows the stress-strain behavior predicted by the numerical model and the analytical softening equation and how they compare with the laboratory direct shear test results from the three sites. It can be observed that the constitutive equation has been well implemented in the numerical model to accurately depict the stress-strain behavior of sensitive clays up to the remolded strength. Additionally, the simulated direct shear test results closely resemble those obtained from laboratory tests, particularly within the laboratory's maximum strain range. These observations suggest that this constitutive model is a reliable tool for predicting the material behavior of sensitive clays under undrained shear loading.

4. Description of the numerical models

Figure 5 illustrates two-dimensional plane strain models for the numerical simulation of the Sainte-Monique (1994), Saint-Jude (2010), and Saint-Luc-de-Vincennes (2016) landslides. The models' dimensions are determined based on the in-situ topography, as detailed in a study by Locat et al. (2013, 2015, 2017) and Tremblay-Auger et al. (2021).

For the Sainte-Monique site (Fig. 5a), the dimensions of the left-hand riverbank in the studied area are 21.7 m in height and 183 m in length. The right-hand embankment, which prevents further sliding, is 11 m high and 57 m long. The length of the slope at the left side of the brook is 37 m with a slope angle of 24°, while the right-hand slope is 25 m in length with a slope angle of 26°. The soil slope comprised a very thin 2 m sandy crust (C1), with a thick sensitive clay layer up to 40 m depth. According to the soil investigation results (Fig. 6a), the undrained shear strength remains relatively constant from a depth of 2 to 10 m, measuring at 25 kPa. Progressing to a depth of 10 to 20 m, an increase from 30 to 50 kPa is observed, resulting in an average overall shear strength of 40 kPa. Notably, a direct shear test at 13.5 m depth revealed a shear strength of 40.5 kPa. The shear strength rises from 50 to 70 kPa between 20 and 25 m depth. In the MPM model, a layer of sensitive clay

Table 1
Input parameters for the numerical simulation of DSS tests.

Parameter	Sainte-Monique	Saint-Jude	Saint-Luc-de-Vincennes
Natural weight of soil, m	16 kN/m^3	16.5kN/m^3	16 kN/m^3
Effective vertical stress, σ_v'	93 kPa	170 kPa	168 kPa
The coefficient of lateral earth pressure at rest, K_0	0.5	0.5	0.5
Shear Modulus, $G=\frac{s_{up}}{\gamma_p}$	4050 kPa	4580 kPa	4583 kPa
Undrained Poisson's ratio, μ	0.45	0.45	0.45
Peak shear strength, s_{up}	40.5 kPa	55 kPa	55 kPa
Peak shear strain, γ_p	0.01	0.014	0.012
Remolded shear strength, s_{ur}	0.736 kPa	0.92 kPa	1.375 kPa
Residual strain, γ_R	1.8	2.4	1.95
Site-specific constant, α	0.95	1.049	0.945
Site-specific constant, β	2.8	2.5	2.9

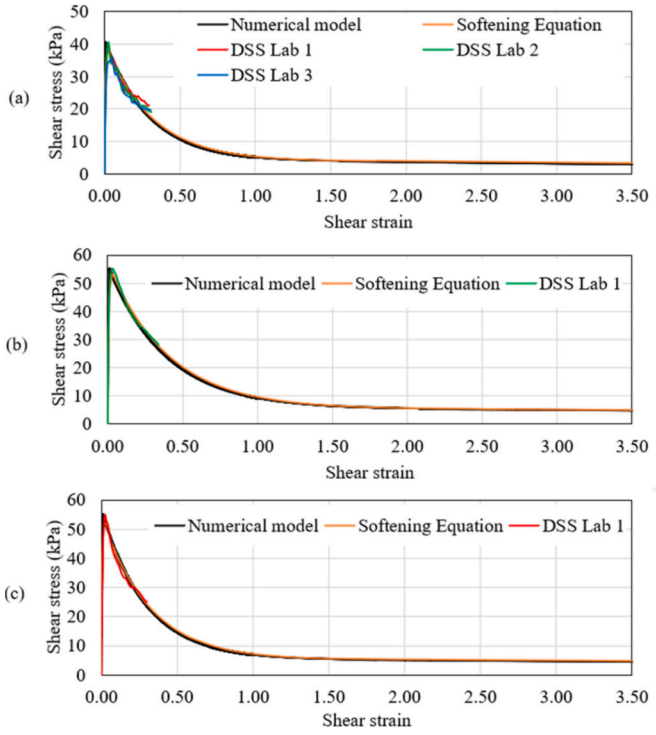


Fig. 4. Stress-strain behavior predicted by the numerical model, analytical softening equation, and DSS laboratory tests for (a) Sainte-Monique, (b) Saint-Jude, and (c) Saint-Luc sites, respectively.

(S1) has been incorporated up to the slope's toe, showcasing a peak shear strength matching that of the Direct Shear Test (DSS) at 40.5 kPa. A 5 m layer of sensitive clay (B1) below the slope's toe is taken to account for the higher shear strength observed in deeper layers. This lower layer has a shear strength of 55 kPa, contributing to a more comprehensive representation of the subsurface conditions. The material properties for the numerical model of the Sainte-Monique site are summarized in Table 2. Note that in all models presented in this work, the sensitive lay layers are modeled with the elastoplastic Mohr-Coulomb model with strain-softening (Eq.2), while other materials involved in the slope failures (e.g., crust layer) are defined with a simple elastic-perfectly plastic Mohr-Coulomb constitutive soil model.

For the Saint-Jude site (Fig. 5b), the slope profile is based on stereophotogrammetry and field investigation data from the report by Locat et al. (2017). The bottom of the river channel is 4.5 m above the lower boundary of the sensitive clay layer. The total height of the slope on the right side of the river is 32.5 m, with a slope angle near the toe of 40°, transitioning to a milder angle ranging between 12 and 20° in the upper part. The ground surface on the right side of the slope crest is almost horizontal. The profile consists of a 4 m thick crust below the ground surface (C2), followed by a 23.5 m thick sensitive clay layer (S2). In the field investigation report, three soil layers with relatively high shear strength were found below the sensitive clay layer and above the bedrock. These layers had a combined thickness of approximately 17 m. For the numerical modeling, these three layers have been simplified as a single soil layer (B2) with a depth of 5 m, as varying the layers' thickness or number of layers did not significantly affect the numerical results (Urmi et al., 2022). The material properties for the numerical simulation of the Saint-Jude site are presented in Table 3. The shear strength parameters of the crust and the till layer are determined by averaging the shear strength data obtained through field investigation (Fig. 6b), corresponding to specific depths of the layers.

For the Saint-Luc-de-Vincennes site, the analyzed slope extends over 320 m, with the left and right embankments measuring 25 m and 10 m in height, respectively (Tremblay-Auger et al., 2021). The geological

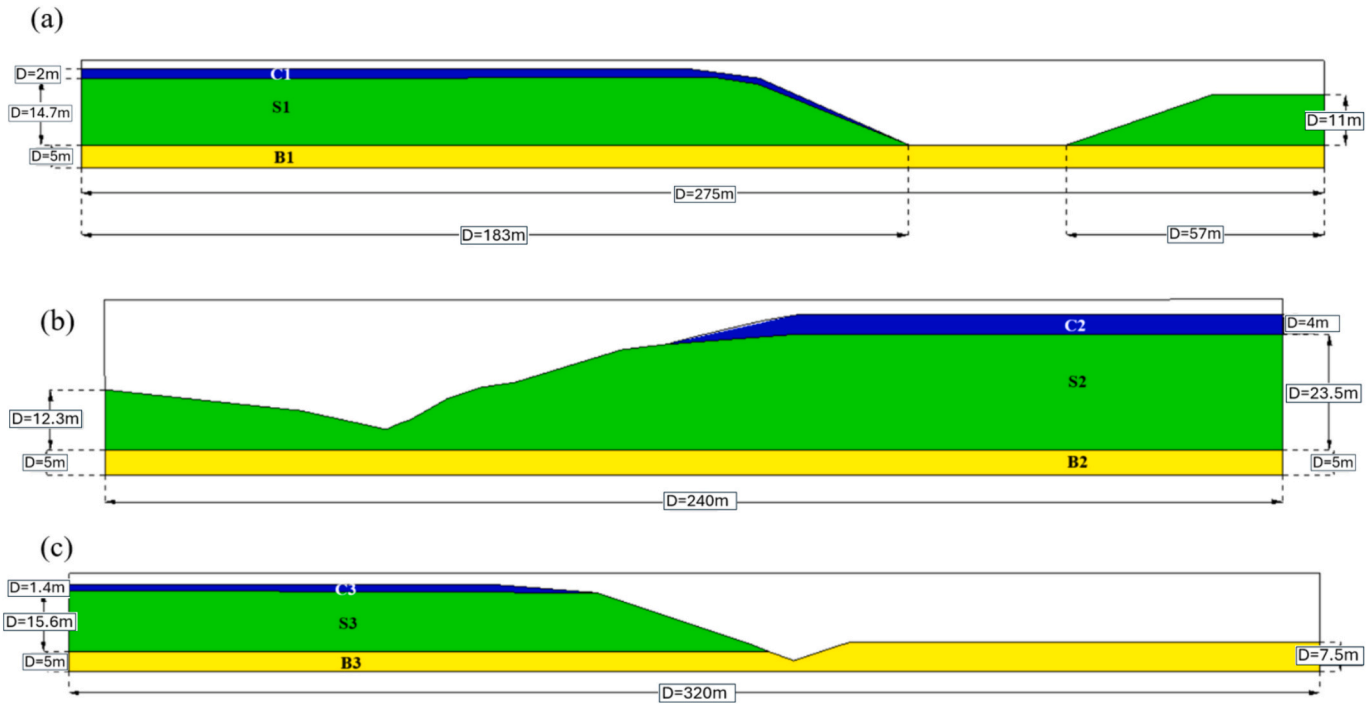


Fig. 5. The geometry of the slope used in the numerical model (a) Sainte-Monique, (b) Saint-Jude, (c) Saint-Luc-de-Vincennes.

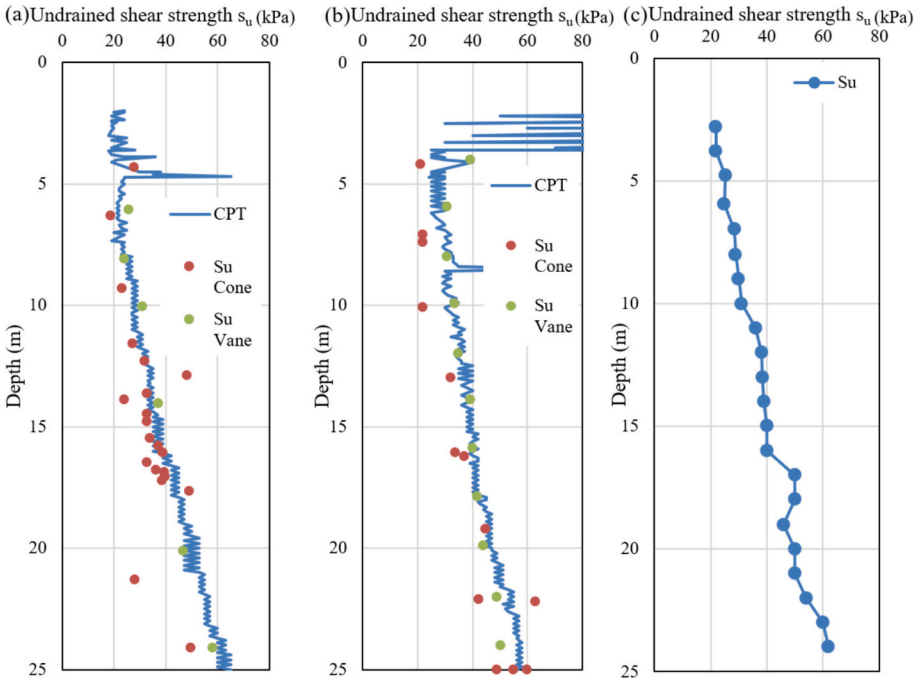


Fig. 6. Depth-wise variation of undrained shear strength; (a) Sainte-Monique, (b) Saint-Jude, (c) Saint-Luc-de-Vincennes.

composition exhibits distinct layers. Starting from the uppermost stratum, a silty-sandy crust layer (C3) persists down to a depth of 1.4 m, succeeded by a sensitive clay unit spanning from 1.4 to 15.5 m (S3), followed by a stiffer sensitive clay layer (B3). In the analysis, two distinct layers of sensitive clay are considered to account for the variation in undrained shear strength with depth (Fig. 6c). For the first layer (S3), a constant peak shear strength of sensitive clay is considered in the model, which is consistent with the DSS test. To represent the sharp increase in shear strength after 15.5 m, a sensitive clay layer (B3) with a higher peak strength has been considered beyond this point. All material parameters of the simulation are summarized in Table 4.

When simulating real-scale landslides, selecting an appropriate mesh size is crucial when dealing with mesh-based continuum numerical methods that incorporate strain-softening features because they result in mesh-dependent strain localization problems and can result in different failure and post-failure predictions (Oliver and Huespe, 2004). While simulating direct shear tests with a mesh size consistent with the soil specimen tested in the lab (laboratory scale) provides accurate stress-strain results, employing the same element size for real-scale landslide simulations becomes computationally very expensive. On the other

Table 2
Input material parameters for the numerical simulation of the Sainte-Monique landslide.

Parameter	Crust (C1)	Sensitive clay layer1 (S1)	Sensitive clay layer2 (B1)
Natural weight of soil, m	18 kN/m^3	17 kN/m^3	17 kN/m^3
The coefficient of lateral earth pressure at rest, K_0	0.5	0.5	0.5
Shear Modulus, G=$\frac{s_{up}}{\gamma_p}$	–	4050 kPa	4050 kPa
Elastic modulus, E	10,000 kPa	–	–
Undrained Poisson's ratio, μ	0.45	0.45	0.45
Peak shear strength, s_{up}	90 kPa	40.5 kPa	55 kPa
Peak shear strain, γ_p	–	0.01	0.01
Remolded shear strength, s_{ur}	–	1 kPa	1 kPa
Residual strain, γ_R	–	1.8	1.8
Site-specific constant, α	–	0.95	0.95
Site-specific constant, β	–	2.8	2.8

Table 3
Input parameters for the numerical simulation of the Saint-Jude landslide.

Parameter	Crust (C2)	Sensitive clay layer (S2)	Stiffer soil layer (B2)
Natural weight of soil, m	18 kN/m^3	16.5 kN/m^3	17 kN/m^3
The coefficient of lateral earth pressure at rest, K_0	0.5	0.5	0.5
Shear Modulus, G=$\frac{s_{up}}{\gamma_p}$	–	4580 kPa	–
Elastic modulus, E	10,000 kPa	–	10,000 kPa
Undrained Poisson's ratio, μ	0.45	0.45	0.45
Peak shear strength, s_{up}	30 kPa	55 kPa	80 kPa
Peak shear strain, γ_p	–	0.014	–
Remolded shear strength, s_{ur}	–	0.92 kPa	–
Residual strain, γ_R	–	2.4	–
Site-specific constant, α	–	1.04	–
Site-specific constant, β	–	2.5	–

Table 4
Input parameters for the numerical simulation of the Saint-Luc-de-Vincennes landslide.

Parameter	Crust (C3)	Sensitive clay layer 1(S3)	Sensitive clay layer 2 (B3)
Natural weight of soil, m	18 kN/m^3	17 kN/m^3	17 kN/m^3
The coefficient of lateral earth pressure at rest, K_0	0.8	0.8	0.8
Shear Modulus, G=$\frac{s_{up}}{\gamma_p}$	–	4583 kPa	4583 kPa
Elastic Modulus, E	10,000 kPa	–	–
Undrained Poisson's ratio, μ	0.45	0.45	0.45
Peak shear strength, s_{up}	25 kPa	55 kPa	70 kPa
Peak shear strain, γ_p	–	0.012	0.012
Remolded shear strength, s_{ur}	–	1.375 kPa	1.375 kPa
Residual strain, γ_R	–	1.95	1.95
Site-specific constant, α	–	0.945	0.945
Site-specific constant, β	–	2.9	2.9

hand, while coarser meshes are computationally more efficient, they can lead to erroneous outcomes if localization problems are not addressed. The smeared crack approach (Rots, 1991) is a simple regularization technique that has been previously used in MPM to address the issue (Alsardi et al., 2021; Soga et al., 2016; Troncone et al., 2022b; Zambrano-Cruzatty and Yerro, 2020). This approach assumes that the total work dissipated by a shear band is equivalent to the fracture energy dissipated in a discrete crack, and consistency in the dissipated work across shear bands of different element sizes ensures consistent results, irrespective of the mesh size employed. This can be achieved by conducting a series of numerical shear tests using various element sizes suitable for field scale modeling. By calibrating large deformation parameters of the softening equation, it is possible to ensure that the area under the stress-displacement curve (dissipated work) is equal for field scale and laboratory scale element sizes. If the dissipated energy is equal in the shearing process, the post-failure movements should be similar per the smeared crack approach.

To achieve this, direct shear tests are simulated for all three sites using a mesh size of 0.5 m, which is the mesh size adopted in the landslide models at the field scale. Both laboratory and field scale direct shear tests provide individual stress-strain curves. The stress-displacement curves for each case are obtained by multiplying the strain values by the element size. The dissipated energy in the shearing process is calculated by integrating the areas under the stress-displacement curves for each element size. To reduce mesh dependency, it is essential to ensure these areas are equal. Since the major part of the stress-displacement curves is exponential, the most straightforward method to adjust the area under the curve is by modifying the exponent of the curve, referred to as the β value. Calibrating the β value, a unique value of β was obtained for each field scale DSS simulation that produces an equal area under the stress-displacement curve obtained from the laboratory scale DSS simulation. The stress-displacement curves employed for the β calibration are depicted in Fig. 7, and the calibrated β values are presented in Table 5.

For all three cases, a single-phase total stress MPM formulation is considered; hence, excess pore pressure is not calculated. It is generally believed that the undrained strain softening is the result of increased pore water pressure in undrained shearing. By implementing the strain-softening behavior, the numerical model captures the response of increased pore water pressure in undrained shearing, simulating the reduction in effective stress and its impact on shearing resistance. The left and right boundaries were fixed horizontally, and the bottom boundary was fixed both horizontally and vertically. The initial stress conditions (Fig. 8) were generated by gravity loading using drained shear strength obtained from site investigation, as presented in Table 6 (Locat et al., 2015, 2017; Tremblay-Auger et al., 2021). As the clay is normally to slightly overconsolidated, the earth pressure coefficient Ko was selected to be 0.5. The effective Poisson's ratio was 0.33, derived from Ko.

The calculation process can be summarized in two main steps follows.

- First, the stresses were initialized by applying gravity loading using drained soil strength parameters (Table 6). An elastoplastic Mohr-Coulomb constitutive soil model without strain softening was used, and slope stability was attained. In this stage, the material behaves mainly elastically, and failure was consistently not reached. The homogeneous local damping in this stage was set to 0.75, given that stress initialization should be a quasi-static problem, and the approach of applying artificial local damping is utilized to enhance numerical convergence speed, as recommended by Ceccato and Simonini (2019).
- In the second calculation step, the failure is triggered by switching the material parameters to be consistent with undrained conditions, as summarized in Tables 2, 3, and 4. A local damping of 0.02 is employed in this stage to facilitate dynamic analysis. The calculation finished when final stability was reached.

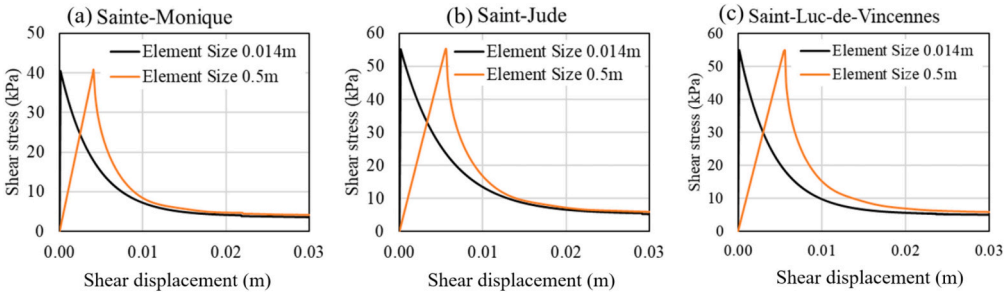


Fig. 7. (a-c): Calibration of β for 0.5 m Element Size (ES).

Table 5
: Calibration of β for 0.5 m element size (ES).

Parameter	Sainte-Monique	Saint-Jude	Saint-Luc-de-Vincennes
For Element Size 0.014 m Site-specific constant, β	2.8	2.5	2.9
For Element Size 0.5 m Site-specific constant, β	105	80	90

5. Results and discussion

5.1. Failure mechanism observed in the simulation compared to the literature and field observations

5.1.1. Sainte-Monique landslide

After the gravity loading (Fig. 9a), the soil is stable with an undrained shear strength of 40 kPa, and the strain starts to localize in a thin shear band as soon as strain softening is activated (Fig. 9b). Failure propagates horizontally through the layer opposite to the river producing a slide bottom "ba"; simultaneously, a slip surface "bc" is formed at an angle of 55° parallel to the slide bottom (Fig. 9c). When stresses along "bc" reach near zero, the soil mass starts to slip, forming a wedge "abc" (Fig. 9c). When "abc" moves forward, the drag of this slip causes a rupture "de" almost parallel to "bc." By this time, the slide bottom has already reached point "d" (Fig. 9d), and a secondary failure surface along "cd" creates a wedge "cde". The wedge "cde" sinks from its original position, creating a graben which leaves a horst at "bcd" (Fig. 9e). Thus, the whole strip "AA'"forms in a retrogressive and discontinuous manner (Fig. 9f) and slips into the river with an almost horizontal (slightly inclined towards the bottom) translational movement. During the movement, tension cracks (or bands) are formed within the wedges, contributing to the formation of horst and graben-like structures. Similarly, strips "BB' "(Fig. 9j) form. During the formation of strip "CC'"(Fig. 9k), the front portion of the moving debris is arrested on the

Table 6
Material parameters for the initial stress condition by gravity loading.

Parameter	C1	S1	B1	C2	S2	B2	C3	S3	B3
Effective cohesion, c' (kPa)	0	3.7	3.7	0	7.7	7.7	2	7.6	7.6
Friction angle, φ' (Degrees)	35	30.6	30.6	35	35	45	30	32	32

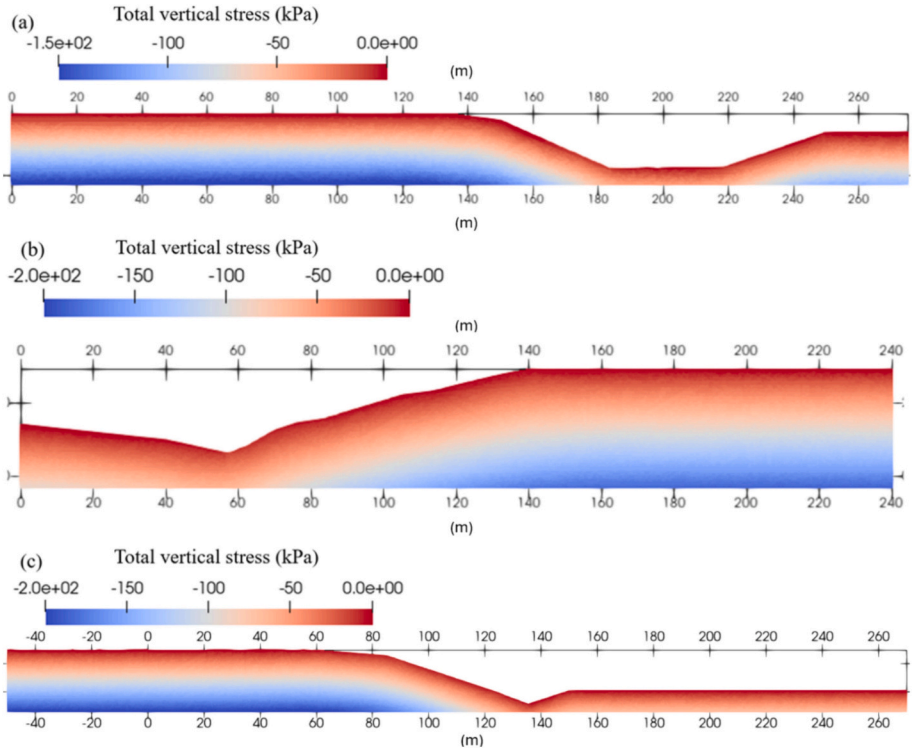


Fig. 8. Total vertical stress of the slope; (a) Sainte-Monique, (b) Saint-Jude, (c) Saint-Luc-de-Vincennes.

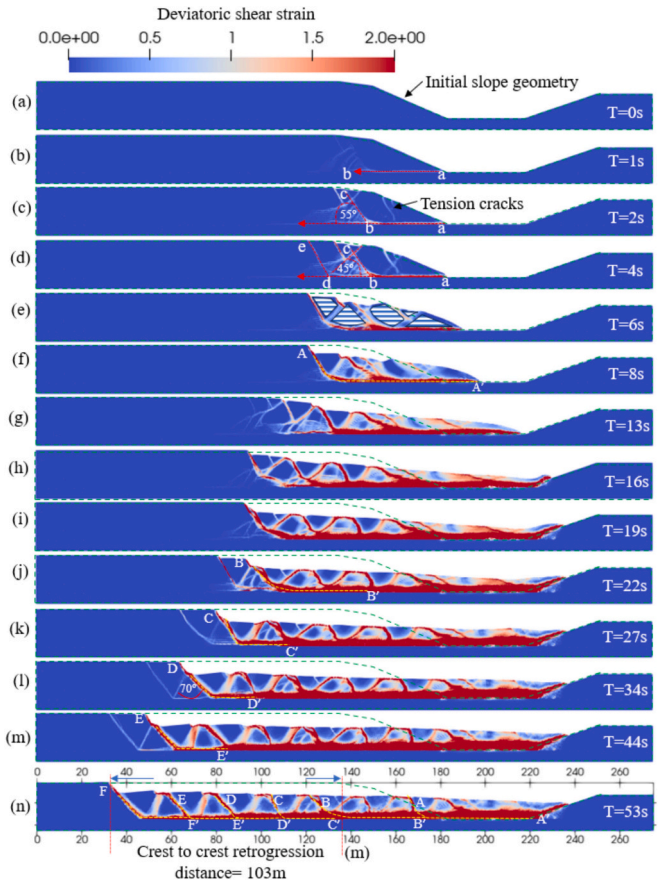


Fig. 9. The failure mechanism of simulated retrogressive Sainte-Monique landslide.

other side of the river, and the landslide movement starts to cease. Then, strips "CC'"(Fig. 9n) and "DD'"(Fig. 9p) are formed. After this point, no more tension cracks are formed, and the formation of horst and grabens is straightforward, with the formation of primary and secondary alternate failure surfaces, as presented in Fig. 9c-d. Similarly, strips "DD'", "EE'", and "FF'"form, each having one horst and one graben (Fig. 9l-n). The landslide completely stops when strip FF' propagates about 4 m from its original position (Fig. 9n). The final retrogression distance is about 103 m from the original position of the crest. The tip angles of the horsts are 70° on average. The failure observed mechanism perfectly complies with the description of the spread mechanism of Odenstad (1951) and Carson (1977), as summarized in Urmi et al. (2023a).

The numerical simulation accurately captured the retrogression observed in the field, resulting in a similar retrogression distance and post-failure geometry (Fig. 10). The crest-to-crest retrogression observed in the field was 100 m, while the numerical model predicted 103 m. A comparison of the simulation's post-failure features with the field observation by Locat et al. (2015) reveals several similarities.

- In the simulation, the landslide debris remained inside the crater, with the opposite side of the brook obstructing the flow and resulting in minimal lateral movement, as observed in the field.
- In the simulation, several ridges were visible with intact material, and some had sharp tips pointing upwards, referred to as horsts, with tip angles mostly around 70°. The field observations noted that the tip angle of the horst closely resembled the active failure angle observed in eastern Canadian clays, approximately 60°. Other blocks within the simulated debris had flat, horizontal tops called grabens, which were not entirely separated and could be connected by vegetation, as observed in the field.
- In the simulation, the soil mass moved horizontally in an active failure mode, consistent with field observations.

5.1.2. *Saint-Jude landslide*

After the gravity loading (Fig. 11a), the slope is stable with an undrained shear strength of 55 kPa, and the strain starts to localize in a thin shear band as soon as strain softening is activated (Fig. 11b). A circular failure surface, "abc" forms from the toe of the slope. When stresses along the shear band reduce, resulting from the large straining and the strain-softening behavior, the clay within becomes fully remolded, and the slope mass starts displacing towards the river in a translational movement (Fig. 11c). As strain concentration intensifies within the shear band, the enclosed soil mass experiences lateral spreading. This lateral expansion leads to strain localization within the central portion of the moving soil, dividing the soil mass into two distinct segmented blocks (Fig. 11d-e). The movement of the liquified debris generates another failure surface, labeled "def," characterized by numerous secondary shear bands. These shear bands fracture the soil mass into segmented blocks. These segmented blocks appear as horst and grabens (highlighted with striped zones in Fig. 11), generally observed in spread failures. This mechanism of spread failure is similar to Mollard and Hughes's, 1973 description of spread failure. The third and final rupture occurs as a wedge "ghi," and the movement of the debris ceases at 32 s after the failure is triggered, with a final crest-to-crest retrogression distance of 70 m, which is within the range of the retrogression distance observed over three different cross-sections on the field (60-75 m). The difference could be related to the cross-section taken to define the geometry of the model. Comparing the two different cross-sections of the field geometry with the numerical model, it is observed that the retrogression distance obtained in the numerical model is 7 m less than the observed retrogression in the field when compared with cross-section BB' (Fig. 12) and 6 m more when compared with the cross-section AA' (Fig. 13). Despite this difference, the post-failure geometry of the simulated slope is very similar to the field observation (Fig. 12). A comparison of the simulation's post-failure features with the field observation by Locat et al. (2017) reveals the following similarities and deviations.

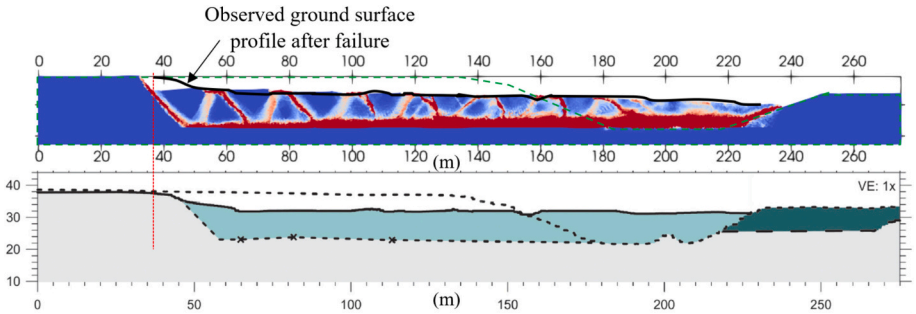


Fig. 10. Comparison of the post-failure geometry between simulated MPM results and field observation by Locat et al. (2015) of Sainte-Monique landslide.

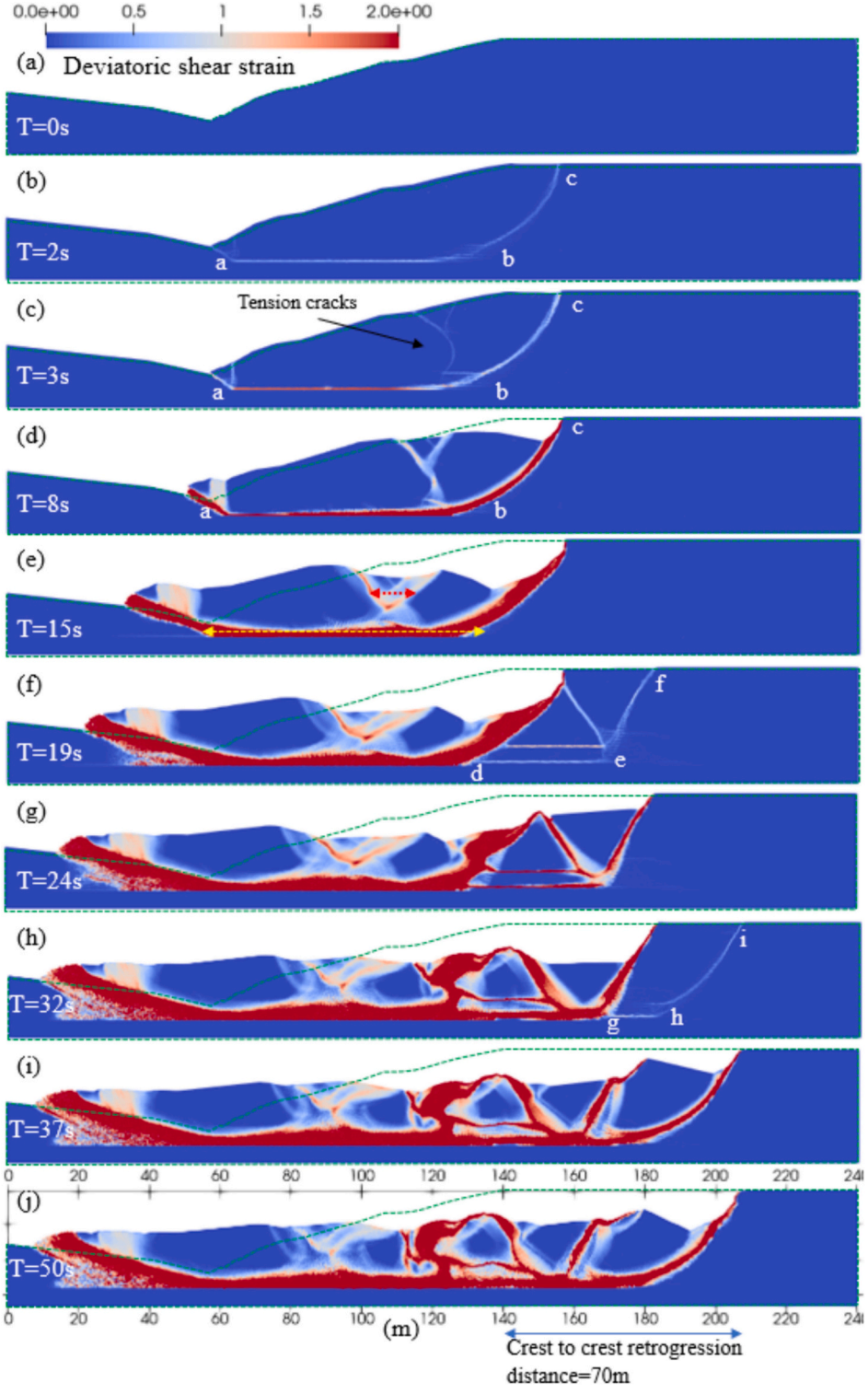


Fig. 11. The failure mechanism of simulated retrogressive Saint-Jude landslide.

- In the field, the failure surface was identified as 2.5 m below the river's level by CPTu and progressed nearly horizontally for 100 m within the undisturbed deposit. In the numerical analysis, the failure surface formed around 3 m beneath the river elevation and propagated roughly 65 m horizontally within the intact deposit.
- In line with the field data, the modeled slope moved to the opposite side of the river, causing minimal disturbance in the debris. Behind it, the soil mass split into multiple segments, adopting horst and graben formations.
- A secondary failure surface, approximately 10 m higher than the primary one, was also identified using in-situ field testing with CPTUs. In contrast, the failure surfaces in the numerical model were at the same elevation.
- The simulated debris movement was mainly translational with minimal or no rotation with a continuous failure surface, as observed in the field.

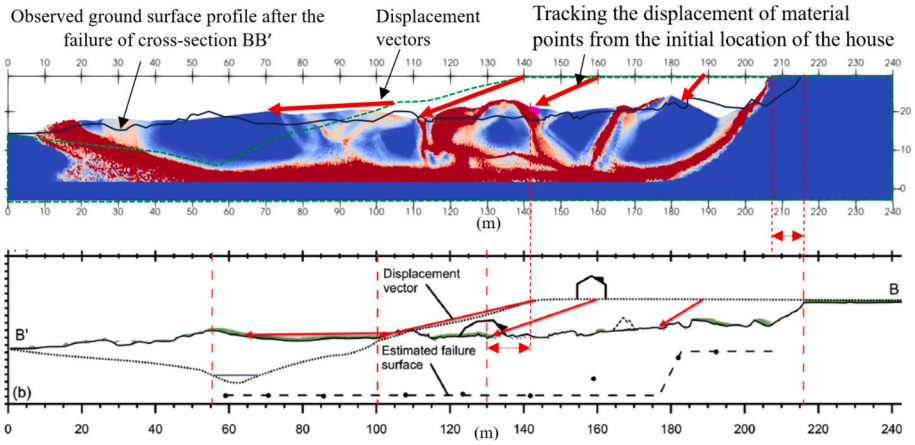


Fig. 12. Comparison of the post-failure geometry of the simulated model and field observation of Saint-Jude landslide of Locat et al., 2017

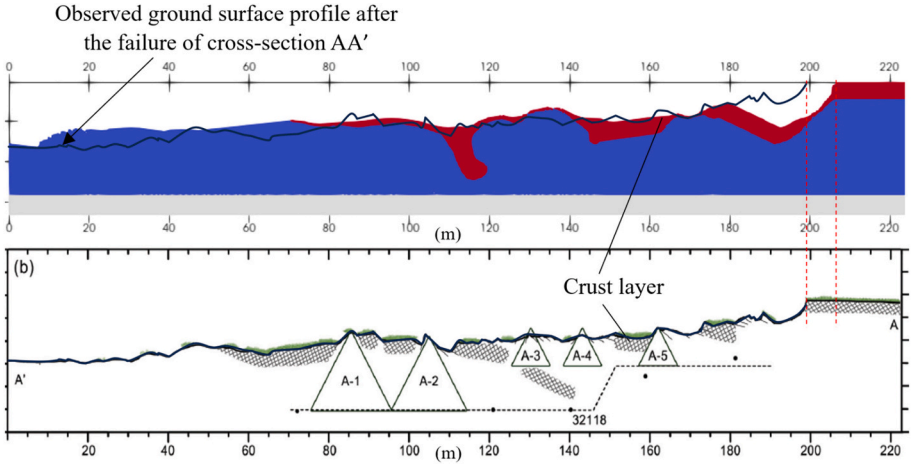


Fig. 13. Comparison of the position of the crust layer in the numerical simulation of this study with Locat et al. (2017)'s post-failure field observations of the Saint-Jude landslide.

- The location of the crust layer after failure observed in both the field and the numerical analysis closely align with each other, as illustrated in Fig. 13.

5.1.3. Saint-Luc-de-Vincennes landslide

After the initial gravity loading as depicted in Fig. 14a, the soil remains stable, exhibiting an undrained shear strength of 55 kPa. Upon activating strain softening, the strain begins to localize within a narrow shear band, as illustrated in Fig. 14b. This shear band formation is analogous to that observed in the Saint-Monique landslide, where tension cracks (or bands) and alternating shear bands lead to the formation of strip AA' (Fig. 14e). This strip then experiences rapid flow (Fig. 14e-l) over the riverbed, resulting in the disappearance of the initially formed horst and graben structures due to this movement (Fig. 14n). The rest of the horst and graben structures are formed in successive strips and remain in the crater because their movement is restricted by the debris left from the flow of the initial strip. Finally, when the slope finally stabilizes, the landslide displays horst and grabens from the final backscarp to a distance of 62 m, designating this region as a spread failure. The remaining portion of the landslide is classified as a flow slide, as depicted in Fig. 14o. Overall, the simulated landslide led to a retrogression of approximately 94 m and had a runout of about 135 m from the initial crater. Typically, in spread failures, the runout is substantially shorter than the retrogression distance. However, this particular failure presented both horst and graben structures, as well as a significant zone of remolded debris, constituting what can be described as a combined failure mode.

As per the numerical simulation, the reason behind the soil blocks within the first strip transforming completely into liquid debris is the lack of geometrical restrictions to block their rapid movement. Notably, the mechanism of shear band formation and its propagation remains consistent between the Sainte-Monique landslide and the present case, apart from the fact that the sliding front ceases to move due to the restriction of the opposite riverbank in the Sainte-Monique case.

In comparison with field observations (Fig. 15), the numerical results present both remarkable similarities and notable differences:

- The simulated landslide initiated with two rotational failure surfaces, as shown in Fig. 13c, which flowed rapidly out of the crater. Following these initial slides, alternate shear bands gave rise to horst and graben configurations, resulting in a composite spread and flow slide failure, mirroring the field observations.
- In the field, the observed failure surface inclined progressively upwards with each retrogression. However, the simulation revealed a consistently horizontal failure surface, positioned approximately 4 m below the rupture plane observed in the field. It could be because of the inhomogeneities or inclination of soil layers that are not captured in the model.
- Although the simulation identified the zone of spread failure nearly at the same distance from the final crater as observed in the field (62 m in the model, 66 m in the field), the model underestimated the final retrogression distance by 18 m. The model's prediction was

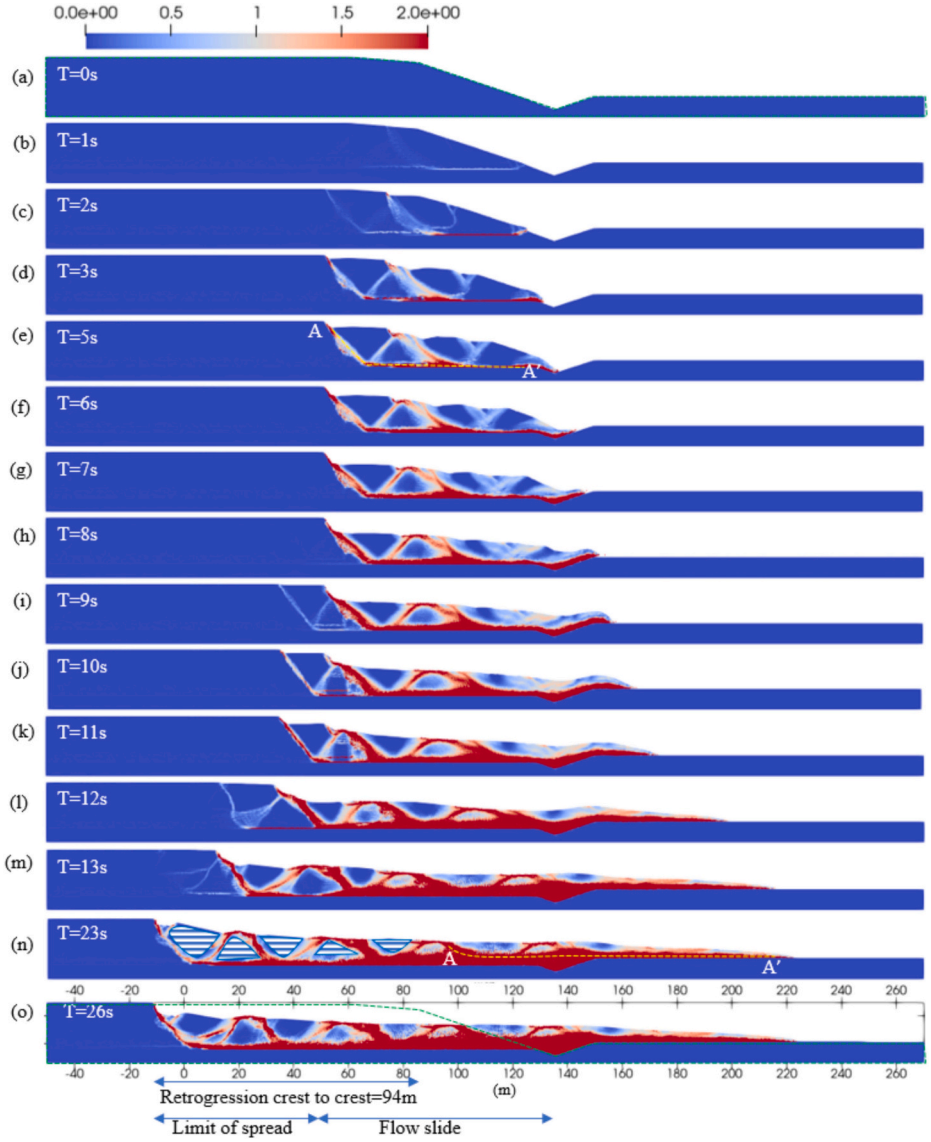


Fig. 14. The failure mechanism of simulated retrogressive Saint-Luc-de-Vincennes landslide.

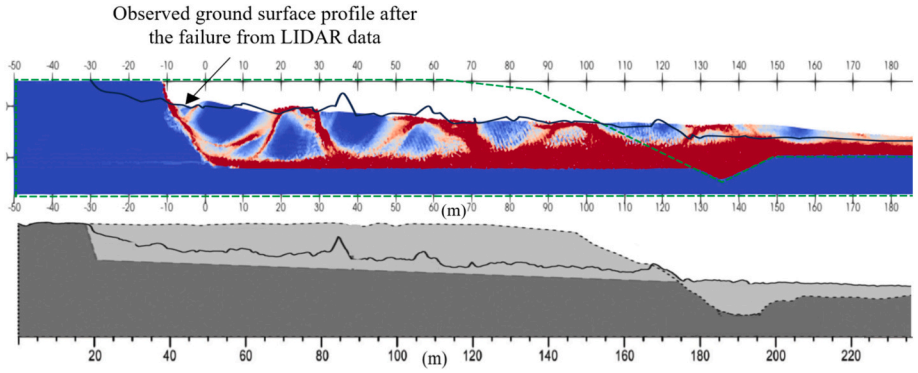


Fig. 15. Comparison of the post-failure geometry of the simulated model and field observation of Saint-Luc-de-Vincennes landslide of Tremblay-Auger et al. (2021).

84% accurate for the retrogression distance and 90% accurate for the spread limit.

5.2. Comparison of the simulation results with other numerical simulations of the same landslide

Tran and Sołowski (2019) and Shan et al. (2021) undertook modeling of the Sainte-Monique landslide. They adopted different

strain-softening equations, integrating them into the Generalized Interpolation MPM framework (GIMPM) and the Particle Finite Element Method (PFEM) with the RITSS technique, respectively. A significant similarity in their approaches was the consideration of the strain rate's effect on shear strength and the depthwise fluctuation in peak shear strength.

Tran and Sołowski (2019) developed a numerical model that predicted retrogression distances equal to the field using remolded undrained shear strengths of 1.6 kPa, 2.5 kPa, and 3 kPa at the failure surface, coupled with strain rate parameters (β) of 0, 0.06, and 0.17, respectively. These cases correspond to sensitivities of 25, 16, and 13.3. Their sensitivity analysis revealed that increasing β necessitates a higher remolded strength and a lower sensitivity to align more closely with field observations. For the base case, they selected a remolded shear strength (s_{ur}) of 1.6 kPa and a sensitivity (S_t) of 25, as this combination most closely matched the real field parameters of Sur = 0.7 kPa and S_t=55. Intriguingly, they also found that choosing β = 0.17, while keeping s_{ur} at 1.6 kPa and S_t at 25, yielded a retrogression distance similar to field observations. This similarity in outcomes across different β values, specifically between β = 0 and β = 0.17, with the same s_{ur} and S_t, was not explicitly explained in their findings, raising questions about the influence of β on model predictions under these conditions. The failure mechanism and post-failure dynamics of their study were similar to this study.

Shan et al.'s (2021) results diverged from several field observations, notably the missing ridges and the complete fragmentation of the soil mass into separate blocks. This current study utilized a more straightforward constitutive model compared to the previous studies. Despite the simplicity, the constitutive model has more effectively captured the post-failure behavior, as demonstrated in Fig. 16. This enhanced behavior might be traced back to the strain-softening equation, derived after scrutinizing the large-deformation stress-strain behavior of multiple sensitive clay sites in eastern Canada (Urmi et al., 2023b). Furthermore, this study's material parameters were directly rooted in laboratory direct shear test data, and no calibration was performed to match landslide simulations with field observations, while Shan et al. (2021) calibrated the soil parameters by doing sensitivity analysis on the large deformation parameter, displacement at remolded shear strength with δ=0.05, 0.1, 0.2 and 0.4 m and selected δ= 0.1 because it yielded to the observed retrogression distance of the field.

On the other hand, Wang et al. (2021), Zhang et al. (2020), and Shen et al. (2024) analyzed the Saint-Jude landslide; Zhang et al. (2020) and Shen et al. (2024) employed linear strain softening within a viscoplastic soil constitutive model, integrating the soil model in the particle finite element and arbitrary Lagrangian finite element frameworks, respectively. Both studies depicted the Saint-Jude landslide as a multiple rotational retrogressive failure, a portrayal that contrasted with actual field observations (Fig. 17b,c). Even so, the post-failure dynamics regarding runout and retrogression from these studies aligned well with this research. Wang et al. (2021) used the same numerical framework as Shen et al. (2024), but the constitutive soil model is more advanced, with an exponential strain softening that accounted for strain rate and variations in the peak shear strength based on depth. While Wang et al. (2021) depicted segmented blocks that formed horst and grabens differently from our study, the similarities are striking (Fig. 17a,d). Additionally, Wang et al. (2021) replicated the post-failure retrogression witnessed in field observations, whereas our model predicted a retrogression shorter by 20 m.

The numerical framework and the constitutive soil model are different in these studies and our study. Still, the key difference in the modeling methodology lies in the evaluation and incorporation of the large deformation parameters (γ_R,γ_r) in the numerical model. Shen et al. (2024) and Zhang et al. (2020) used a visco-elastic Tresca model that considers linear strain softening. The difficulty with considering a linear softening behavior has been discussed in Urmi et al. (2023a, 2023b). It demonstrates that a realistic outcome considering linear behavior is only possible when the energy dissipated by the remolding process in both the actual non-linear behavior and assumed linear behavior is equal. As per fracture mechanics, if the area under the stress-strain curve in both cases is equal, that represents similar strain-softening behavior. It is only possible when, in linear strength decrease, the over-estimated area under the strain softening curve is compensated by the underestimated area. Without the complete curve up to the point of remolding, uncertainty remains with the selection of the large deformation parameter (residual shear strain) from laboratory shear test data. Both Shen et al. (2024) and Zhang et al. (2020) stated that the value of residual shear strain is determined from the strain-softening pattern of Canadian sensitive clays as per Quinn et al. (2011)'s study. Both studies considered the value of residual shear strain as 1.6 (160%) without specifying the detailed methodology for obtaining it from Quinn et al.'s (2011) data or whether any sensitivity analysis has been done to obtain the large deformation parameter that produces simulation result close to the field observation. The focus of this study is to reduce the uncertainty in determining the residual shear strain by providing a clear methodology for obtaining the residual and remolded strain (γ_R,γ_r) in eq. 2 so that the complete strain-softening curve is readily available to provide the input parameters for effective prediction of the post-failure behavior.

The lack of an appropriate methodology for gauging the large deformation characteristics inherent to sensitive clays required the prior studies to focus on retrospective examinations of past landslides. This generally involved iterative tweaks to large deformation parameters to better fit observed field behavior. Therefore, in contrast to our research, earlier studies discussed above did not forecast the potential extent of either retreat or advancement in sensitive clay landslides. However, it is

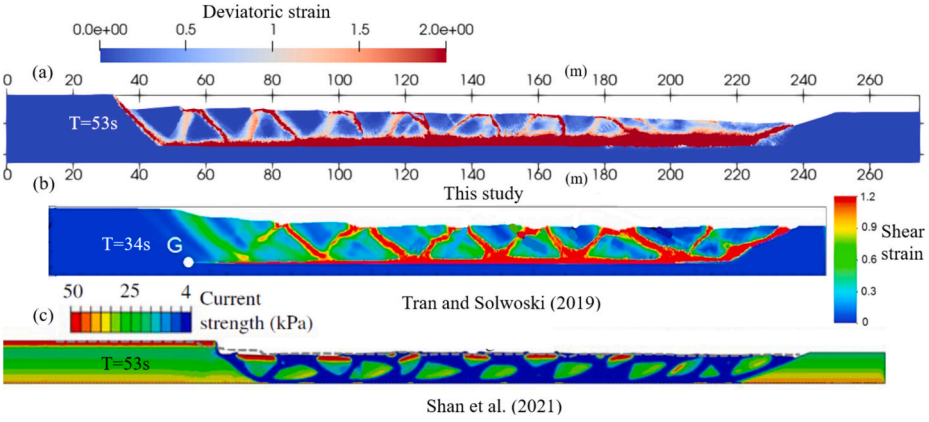


Fig. 16. Comparison of the post-failure geometry of the simulated model of this study and other numerical simulations of the Sainte-Monique landslide; (a) This study; (b) Tran and Sołowski (2019); (c) Shan et al. (2021).

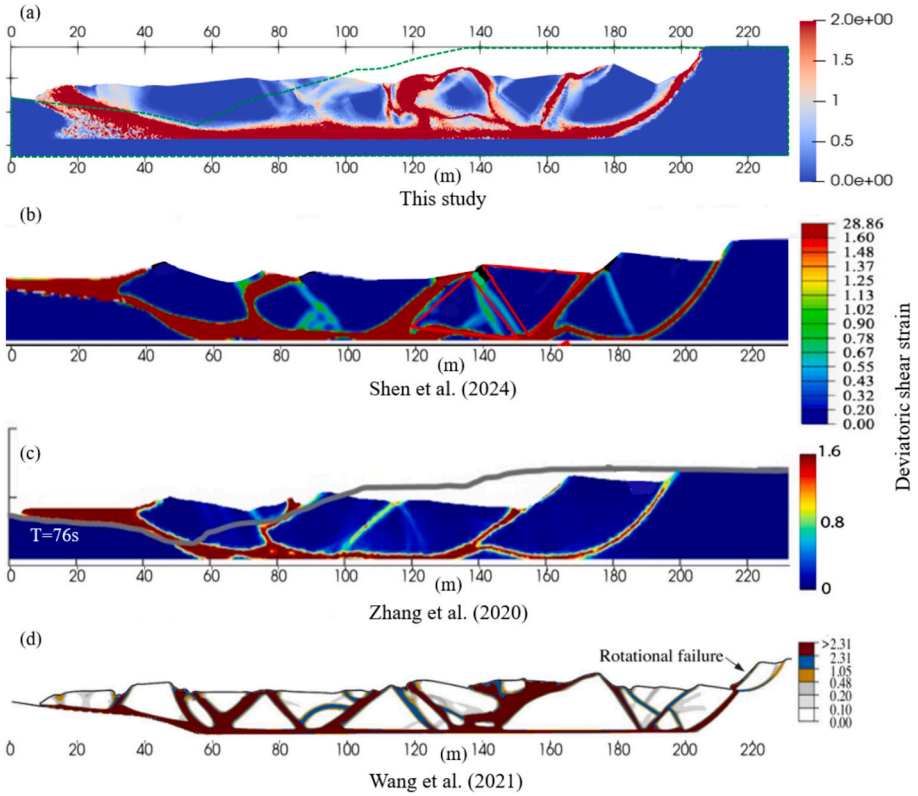


Fig. 17. Comparison of the post-failure geometry of the simulated model of this study and other numerical simulations of the Saint-Jude landslide; (a) this study, (b) Shen et al. (2024), (c) Zhang et al. (2020), and (d) Wang et al. (2021).

important to emphasize that since the depthwise variation of peak shear strength was not considered, the failure surface emerged directly above the point where the deeper layer was characterized by greater strength in the numerical model. Although the soil layer selection was based on field investigation reports and engineering judgment, studies that specifically addressed variations in peak shear strength depicted a continuous sensitive clay layer where the failure surface naturally formed at its weakest point, providing a more realistic representation.

This work introduces a novel constitutive model specifically designed to capture the large strain behavior of sensitive clays. This model allows for effective prediction of clay behavior under stress, eliminating the need for rigorous and often time-consuming back analysis typically required in such studies. The primary advantage of the proposed constitutive soil model lies in its simplicity, as it only requires input parameters that can be easily obtained from basic soil tests, making it both practical and accessible for widespread use. The methodology detailed in this paper is particularly useful for predicting the post-failure runout and retrogression of future landslides in sensitive clays, which are common in the landslide-prone areas of eastern Canada. By accurately simulating these post-failure behaviors, the model helps in understanding and anticipating the movement of remolded clay after a landslide event. This ability to predict such movements is crucial for the development of effective hazard maps and risk assessments. By providing reliable data and predictions, the model helps prepare for potential landslides, enhances community safety and resilience, and offers a valuable tool for mitigating impacts and improving structural stability on sensitive clay deposits.

6. Conclusion

- This research introduced a novel and straightforward strain-softening constitutive soil model for sensitive clays within the framework of the material point method. One can derive the large deformation parameters (γ_R, γ_r) of sensitive clay for specific sites from geotechnical investigation reports. Furthermore, laboratory test outcomes adequately supply the model's necessary input parameters. What sets this constitutive model apart is its reliance on site-specific parameters α, β, residual stress s_{uR}, residual strain γ_R, and remolded shear strain γ_r, all attainable from laboratory and field investigation data. By leveraging the large deformation parameters (γ_R, γ_r) acquired from geotechnical investigation data, this model facilitates a more realistic prediction. A parameter in the model can be calibrated to reduce mesh dependency on the results, bridging the gap between laboratory tests and larger mesh element sizes employed in real-world scenarios through the smeared crack approach.
- Three different Canadian landslides in sensitive clay have been numerically modeled, i.e., Sainte-Monique, Saint-Jude, and Saint-Luc-de-Vincennes landslides. The numerical findings underscore the feasibility of predicting most features in retrogressive failures. The landslides modeled in this research demonstrate the formation and advancement of shear bands consistent with field observations. Moreover, the numerical simulations validate prior characterizations of spread failure mechanisms in the existing literature.
- The Saint-Luc-de-Vincennes landslide is a composite spread and flow type that was not attempted to model numerically before this study. The proposed model has been able to predict the post-failure movement of a composite landslide with around 90% accuracy.
- While the current constitutive model prizes simplicity, there is potential for further refinement. This enhancement could consider pivotal elements currently overlooked, including depth-dependent variations in peak shear strength, strain rate effects, and soil's long-term creep behavior. Incorporating these aspects would elevate the model's precision and range of application.

CRediT authorship contribution statement

Zinan Ara Urmi: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Alba Yerro:** Writing – review & editing, Supervision, Software, Methodology. **Ali Saeidi:** Writing – review & editing, Supervision, Methodology, Funding acquisition. **Rama Vara Prasad Chavali:** Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Ali Saeidi reports financial support was provided by Natural Sciences and Engineering Research Council of Canada (Grant ID: NSERC- 950-232724). Ali Saeidi reports financial support was provided by Hydro-Québec (Grant ID: RDCPJ 521771–17). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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